extremely brief burst of astoundingly rapid expansion during the universe's earliest moments (in this approach, the size of the universe increased by a factor larger than a million trillion trillion in less than a millionth of a trillionth of a trillionth of a second). As will become clear, this stupendous growth of the young universe goes a long way toward filling in the gaps left by the big bang model—of explaining the shape of space and the uniformity of the microwave radiation, and also of suggesting why the early universe might have been highly ordered—thus providing significant progress toward explaining both astronomical observations and the arrow of time we all experience (Chapter 11).

Yet, despite these mounting successes, for two decades inflationary cosmology has been harboring its own embarrassing secret. Like the standard big bang theory it modified, inflationary cosmology rests on the equations Einstein discovered with his general theory of relativity. Although volumes of research articles attest to the power of Einstein's equations to accurately describe large and massive objects, physicists have long known that an accurate theoretical analysis of small objects—such as the observable universe when it was a mere fraction of a second old— requires the use of quantum mechanics. The problem, though, is that when the equations of general relativity commingle with those of quantum mechanics, the result is disastrous. The equations break down entirely, and this prevents us from determining how the universe was born and whether at its birth it realized the conditions necessary to explain time's arrow.

It's not an overstatement to describe this situation as a theoretician's nightmare: the absence of mathematical tools with which to analyze a vital realm that lies beyond experimental accessibility. And since space and time are so thoroughly entwined with this particular inaccessible realm—the origin of the universe—understanding space and time fully requires us to find equations that can cope with the extreme conditions of huge density, energy, and temperature characteristic of the universe's earliest moments. This is an absolutely essential goal, and one that many physicists believe requires developing a so-called *unified theory.*

## Unified Reality